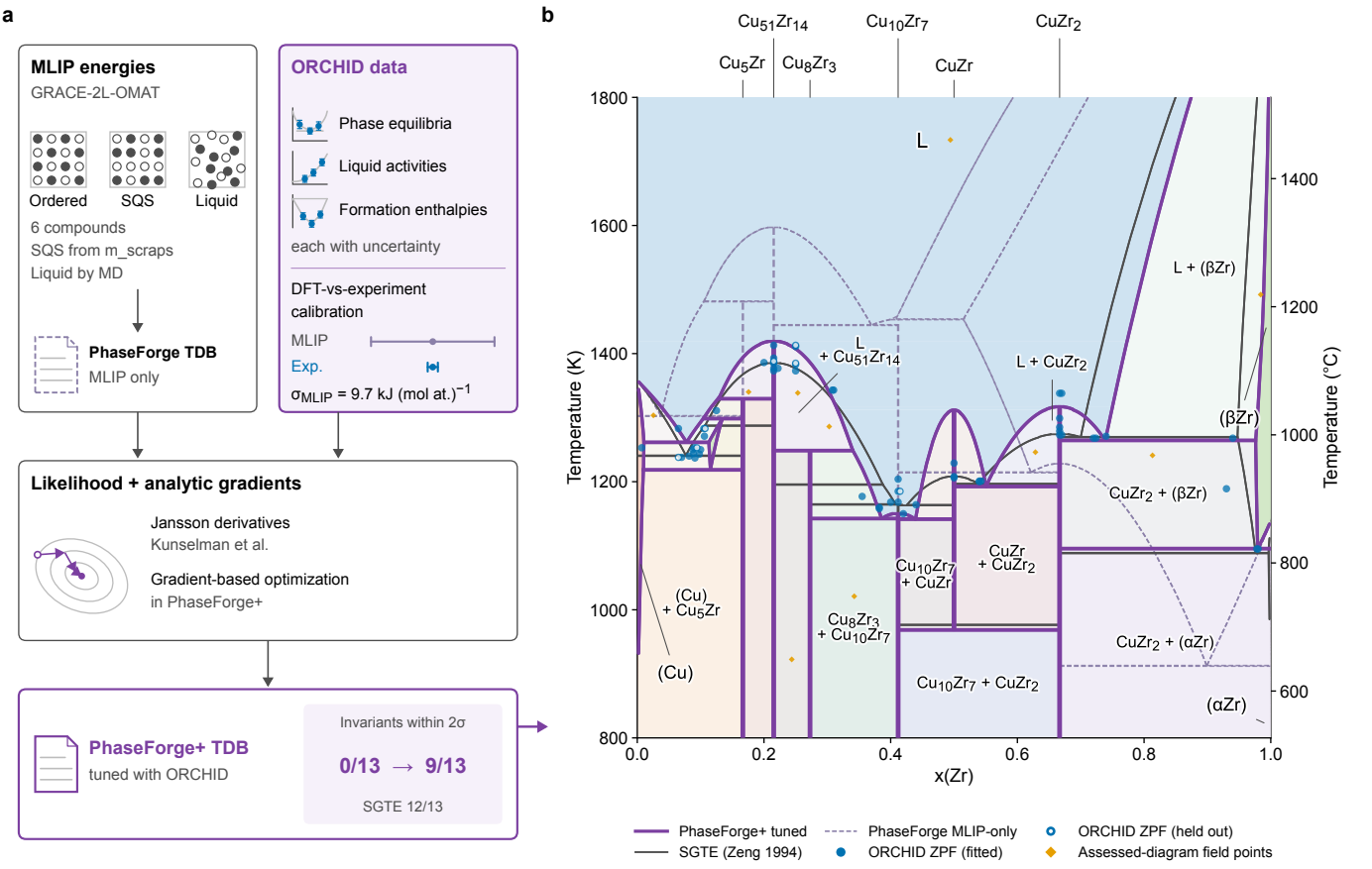

a
MLIP energies
GRACE-2L-OMAT
Ordered
SQS
Liquid
6 compounds
SQS from m_scraps
Liquid by MD
PhaseForge TDB
MLIP only
ORCHID data
Phase equilibria
Liquid activities
Formation enthalpies
each with uncertainty
DFT-vs-experiment calibration
MLIP
Exp.
σMLIP = 9.7 kJ (mol at.)−1
Likelihood + analytic gradients
Jansson derivatives
Kunselman et al.
Gradient-based optimization
in PhaseForge+
PhaseForge+ TDB
tuned with ORCHID
Invariants within 2σ
0/13 → 9/13
SGTE 12/13
b
Cu51Zr14
Cu10Zr7
CuZr2
Cu5Zr
Cu8Zr3
CuZr
L
L + (βZr)
L + Cu51Zr14
L + CuZr2
(βZr)
CuZr2 + (βZr)
Cu10Zr7 + CuZr
CuZr + CuZr2
(Cu) + Cu5Zr
Cu8Zr3 + Cu10Zr7
CuZr2 + (αZr)
(Cu)
Cu10Zr7 + CuZr2
(αZr)
Temperature (K)
Temperature (°C)
x(Zr)
PhaseForge+ tuned
SGTE (Zeng 1994)
PhaseForge MLIP-only
ORCHID ZPF (fitted)
ORCHID ZPF (held out)
Assessed-diagram field points